

Pre-clinical evaluation of *N*-acetylcysteine reveals side effects in the *mdx* mouse model of Duchenne muscular dystrophy

Gavin J. Pinniger¹ , Jessica R. Terrill^{1,2} , Evanna B. Assan¹, Miranda D. Grounds¹  and Peter G. Arthur² 

¹School of Human Sciences, the University of Western Australia, Crawley, Western Australia, Australia

²School of Molecular Sciences, the University of Western Australia, Perth, Western Australia, Australia

Key points

- Duchenne muscular dystrophy (DMD) is a fatal muscle wasting disease associated with increased inflammation and oxidative stress.
- The antioxidant *N*-acetylcysteine (NAC) has been proposed as a therapeutic intervention for DMD boys, but potential adverse effects of NAC have not been widely investigated.
- We used young (6 weeks old) growing *mdx* mice to investigate the capacity of NAC supplementation (2% in drinking water for 6 weeks) to improve dystrophic muscle function and to explore broader systemic effects of NAC treatment.
- NAC treatment improved normalised measures of muscle function, and decreased inflammation and oxidative stress, but significantly reduced body weight gain, muscle weight and liver weight.
- Unexpected significant adverse effects of NAC on body and muscle weights indicate that interpretation of muscle function based on normalised force measures should be made with caution and careful consideration is needed when proposing the use of NAC as a therapeutic treatment for young DMD boys.

Abstract Duchenne muscular dystrophy (DMD) is a fatal X-linked muscle wasting disease characterised by severe muscle weakness, necrosis, inflammation and oxidative stress. The antioxidant *N*-acetylcysteine (NAC) has been proposed as a potential therapeutic intervention for DMD boys. We investigated the capacity of NAC to improve dystrophic muscle function in the *mdx* mouse model of DMD. Young (6 weeks old) *mdx* and non-dystrophic C57 mice receiving 2% NAC in drinking water for 6 weeks were compared with untreated mice. Grip strength and body weight were measured weekly, before the 12 week old mice were anaesthetised and extensor digitorum longus (EDL) muscles were excised for functional analysis and tissues were sampled for biochemical analyses. Compared to untreated mice, the mean (SD) normalised grip strength was significantly greater in NAC-treated *mdx* [3.13 (0.58) vs 4.87 (0.78) g body weight (bw)⁻¹; $P < 0.001$] and C57 mice [3.90 (0.32) vs 5.32 (0.60) gbw⁻¹; $P < 0.001$]. Maximum specific force was significantly greater in NAC-treated *mdx* muscles [9.80 (2.27) vs 13.07 (3.37) N cm⁻²; $P = 0.038$]. Increased force in *mdx* mice was associated with reduced thiol oxidation and inflammation in fast muscles, and increased citrate synthase activity in slow muscle. Importantly, NAC significantly impaired body weight gain in both strains of young growing mice, and reduced liver weight in C57 mice and muscle weight in *mdx* mice. These potentially adverse effects of NAC emphasise the need for caution when interpreting improvements in muscle function based on normalised force measures, and that careful consideration be given to these effects when proposing NAC as a potential treatment for young DMD boys.

(Received 2 March 2017; accepted after revision 30 August 2017; first published online 9 September 2017)

Corresponding author G. Pinniger: School of Human Sciences, University of Western Australia, 35 Stirling Highway, Crawley, Western Australia, 6009, Australia. Email: gavin.pinniger@uwa.edu.au

Abbreviations 2ME, 2-mercaptoethanol; APF, 2-[6-(4-aminophenoxy)-3-oxo-3H-xanthen-9-yl]benzoic acid; CD, cysteine deoxygenase; CSA, cross sectional area; CSD, cysteine sulfinate decarboxylase; DC, detergent compatible; DMD, Duchenne muscular dystrophy; EDL, extensor digitorum longus; FI, fatigue index; FLM, BODIPY FL-*N*-(2-aminoethyl) maleimide; GAPDH, glyceraldehyde 3-phosphate dehydrogenase; GSH, glutathione; L_0 , optimal fibre length; L_o , optimal muscle length; MPO, myeloperoxidase; NAC, *N*-acetylcysteine; OPA, *o*-phthalaldehyde; OTC, 1-2-oxothiazolidine-4-carboxylate; RONS, reactive oxygen and nitrogen species; TCA, trichloroacetic acid; Texas red, Texas Red C2-maleimide.

Introduction

Duchenne muscular dystrophy (DMD) is a fatal X-linked recessive muscle disease arising from mutations in the dystrophin gene that result in the absence of a functional dystrophin protein that is closely associated with the sarcolemma (myofibre cell membrane) and the dystrophin–glycoprotein complex (reviewed by Emery, 2002; Bushby *et al.* 2010). The dystrophin–glycoprotein complex spans the sarcolemma, transfers contractile force from the muscle cytoskeleton to the extracellular matrix, and provides the mechanical stability of the sarcolemma and mechanotransduction signalling (Rando, 2001). The absence of functional dystrophin protein in DMD boys increases susceptibility to repeated bouts of myofibre damage and necrosis, with increasing fibrosis that impairs muscle regeneration. The progressive loss of muscle mass and function ultimately results in premature death from respiratory or cardiac failure (Petrof *et al.* 1993; Grounds, 2008; Kharraz *et al.* 2014). Corticosteroids remain the only standard therapeutic treatment for DMD (Matthews *et al.* 2016; Wood *et al.* 2016), and although they produce modest improvements in muscle strength and function, the numerous side-effects associated with long-term steroid treatment warrant the investigation of alternative therapies (Miyatake *et al.* 2016).

The precise cellular mechanisms leading to severe muscle weakness and wasting in DMD remain unknown. However, the lack of dystrophin has been associated with altered membrane permeability and the disruption of Ca^{2+} homeostasis, invasion of inflammatory cells including neutrophils and macrophages, and the elevated production of reactive oxygen and nitrogen species (RONS) and subsequent oxidative stress (Whitehead *et al.* 2006; Arthur *et al.* 2008; Jackson, 2008; Shkryl *et al.* 2009). RONS, including hydroxyl radicals, can cause irreversible damage to proteins, membrane lipids and DNA, whereas reactive oxygen species (ROS) such as hydrogen peroxide (H_2O_2) and hypochlorous acid (HOCl) can cause reversible oxidation of protein thiols and affect the function of many intracellular proteins (Arthur *et al.* 2008). ROS-induced thiol oxidation is one of the proposed mechanisms contributing to contractile dysfunction and

subsequent muscle weakness in DMD (Crowder & Cooke, 1984; Ferreira & Reid, 2008; Jackson, 2008; Terrill *et al.* 2013*b*), possibly via altered calcium homeostasis and/or impaired myofilament force production (Andrade *et al.* 1998, 2001; Ferreira & Reid, 2008; Jackson, 2008). Elevated RONS production has been associated with increased protein thiol oxidation in both *mdx* mice (Hauser *et al.* 1995) and DMD patients (Haycock *et al.* 1996). Therefore, minimising oxidative stress is a promising approach to improve muscle function in DMD.

Antioxidant administration has been widely evaluated as a potential therapeutic treatment for DMD, mainly using *mdx* mice for preliminary preclinical testing (Whitehead *et al.* 2008; Terrill *et al.* 2012, 2013*a*; de Senzi Moraes Pinto *et al.* 2013; Horvath *et al.* 2016). However, the choice of antioxidant treatment is complicated by the many different types of RONS and the numerous potential sources of RONS production. The use of indirect and/or non-specific antioxidants may hamper the translation of promising results from *in vitro* studies to positive outcomes using *in vivo* preclinical studies (Kim *et al.* 2013).

N-acetylcysteine (NAC) is a widely tested inexpensive antioxidant that is approved for use in humans, making it an attractive potential therapeutic treatment for DMD boys. Oral NAC supplements increase resistance to fatigue in endurance-trained athletes (McKenna *et al.* 2006) with similar improvements in fatigue resistance reported for both *in situ* (Shindoh *et al.* 1990) and *in vitro* (Mishima *et al.* 2005) studies of murine diaphragm muscle. The antioxidant properties of NAC may occur directly through scavenging of RONS including the myeloperoxidase-derived oxidant HOCl and, to a lesser extent, H_2O_2 (Aruoma *et al.* 1989), or indirectly as a precursor of L-cysteine that is required for the synthesis of the antioxidant glutathione (GSH) (Medved *et al.* 2004). Oral supplementation with NAC increases cysteine and GSH in the liver, plasma and skeletal muscle (Medved *et al.* 2004; Dilger & Baker, 2007) and reduces oxidative stress and protein thiol oxidation in skeletal muscles (Medved *et al.* 2004; Matuszczak *et al.* 2005; Terrill *et al.* 2012) of humans and mice.

Several preclinical studies using *mdx* mice have evaluated the efficacy of NAC administration to reduce

the dystrophic pathology in both skeletal (Whitehead *et al.* 2008; Terrill *et al.* 2012; de Senzi Moraes Pinto *et al.* 2013) and cardiac muscle (Williams & Allen, 2007; Fauconnier *et al.* 2010). Acute incubation with high concentrations of NAC (20 mM) *ex vivo* in isolated extensor digitorum longus (EDL) muscles of *mdx* mice reduced the severity of stretch-induced muscle damage (Whitehead *et al.* 2008). Oral supplementation of NAC in drinking water significantly reduced the number of centrally nucleated myofibres (a marker of muscle regeneration in response to necrosis), reduced activation of NF- κ B signalling (a marker of inflammation) in sedentary *mdx* mice (Whitehead *et al.* 2008) and also reduced myofibre necrosis and protein thiol oxidation after treadmill exercise in *mdx* mice (Terrill *et al.* 2012). Despite these promising findings, it has not been established that oral supplementation with NAC prevents loss of muscle strength through *in vivo* or *ex vivo* evaluation of contractile function in *mdx* mice. Furthermore, few studies examining the efficacy of NAC as a therapeutic treatment for DMD have considered the potential adverse systemic effects of orally administering cysteine donors such as NAC.

The primary aim of this study was to establish the capacity of NAC to improve muscle strength of young dystrophic *mdx* (compared with normal) mice using *in vivo* and *ex vivo* measures of contractile function; the secondary aim was to further investigate the systemic effects of NAC treatment.

Methods

Ethical approval

All animal experiments were approved by the Animal Ethics committee at the University of Western Australia and were conducted in accordance with the guidelines of the National Health and Medical Research Council Code of practice for the care and use of animals for scientific purposes (2004), and the Animal Welfare act of Western Australia (2002). Experiments also comply with the ethical principles outlined by Grundy (2015). Experiments were performed on male dystrophic *mdx* (C57Bl/10ScSn^{*mdx/mdx*}) and male non-dystrophic normal wild type control C57Bl/10ScSn (C57) mice supplied by the Animal Resources Centre (Perth, WA, Australia). NAC treatment began at 6 weeks of age, and all mice were sampled at 12 weeks of age. The age of mice and duration of treatment were chosen as they reflect a period when the acute severe phase of myofibre necrosis experienced by *mdx* mice has subsided yet they continue to grow, thus reflecting the growth phase of adolescent DMD boys (Grounds, 2008). Mice were maintained in a 12 h light/dark environment at 20–25°C and supplied with food and water *ad libitum*.

NAC treatment and *in vivo* measurements

NAC (Sigma Aldrich, Sydney, NSW, Australia) was dissolved in acidified water (pH 2.5) and administered as 2% NAC in drinking water for 6 weeks. The four experimental groups ($n = 8$ mice per group) were: (i) *mdx* NAC treated; (ii) *mdx* untreated; (iii) non-dystrophic C57 NAC treated; and (iv) non-dystrophic C57 untreated.

Forelimb grip strength was measured following procedures outlined in the TREAT-NMD recommended standard protocol 'Use of grip strength meter to assess limb strength of *mdx* mice – M.2.2-001', available at <http://www.treat-nmd.eu/research/preclinical/SOPs/>. Briefly, mice grasped a metal bar attached to a force transducer (Columbus Instruments, Columbus, OH, USA) and were gently pulled by the tail. The peak force achieved when the grip was broken was recorded. We recorded five consecutive grip strength measurements for each animal, and averaged the three highest values for further analysis. Grip strength measurements were performed by the same person and data were recorded in grams and also normalised to body weight. Body weight and grip strength of all mice were recorded weekly during the 6 week experiment.

In vitro measures of skeletal muscle (EDL) contractile function

After the 6 week treatment, mice (aged 12 weeks) were anaesthetised (i.p. sodium pentobarbitone; 40 mg kg⁻¹) and the EDL muscle was removed for *in vitro* measurements of contractile function. After removal of the EDL muscle, mice were killed with an overdose of sodium pentobarbitone (i.p.; 100 mg kg⁻¹) and death was confirmed by permanent cessation of respiration, heart beat and reflexes. Contralateral hind limb muscles were removed and snap frozen in liquid nitrogen for biochemical assays. The EDL was mounted in an *in vitro* muscle test system (1200A Intact Muscle Test System, Aurora Scientific, ON, Canada) containing mammalian Ringer solution [NaCl (121 mM); KCl (5.4 mM); MgSO₄·7H₂O (1.2 mM); NaHCO₃ (25 mM); Hepes (5 mM); glucose (11.5 mM) and CaCl₂ (2.5 mM)], continuously bubbled with carbogen (5% CO₂ in O₂, BOC, Western Australia) at 25°C. Muscles were stimulated with 0.2 ms supramaximal square wave pulses via two platinum electrodes parallel to the muscle. Muscle length was manually adjusted until the maximum twitch force was recorded, which was designated optimal muscle length (L_0). The maximum twitch response was analysed for peak twitch force, time to peak twitch force, twitch half-relaxation time and maximum rate of force development.

The force-frequency relationship was evaluated by recording the force responses at stimulation frequencies of

10, 20, 30, 40, 60, 80, 100, 120 and 150 Hz. Each stimulus was separated by a 2 min interval to avoid inducing muscle fatigue. The peak tetanic force was recorded as the stimulation that produced the greatest force throughout the experiment. The EDL was then subjected to a fatiguing protocol consisting of repeated tetanic contractions for 500 ms at 120 Hz. Stimulation was delivered once every 5 s for a total period of 4 min. Resistance to fatigue was evaluated from the fatigue index (FI), the maximum force produced in the final fatiguing contraction as a percentage of the pre-fatigue force. The rate of recovery after fatigue was monitored by recording tetanic contractions at 120 Hz at 1, 5, 10, 15, 20 and 30 min after the fatiguing protocol.

Muscle cross sectional area (CSA) was determined by dividing the wet muscle mass by the product of the optimal fibre length (L_f) and the density of mammalian skeletal muscle (1.06 mg mm^{-3}) (Mendez & Keys, 1960). L_f was calculated using a predetermined L_f to L_o ratio of 0.45 in EDL muscles (Brooks & Faulkner, 1988). Specific force (N cm^{-2}) was calculated by dividing the muscle force (N) by the CSA.

Citrate synthase analysis (to measure mitochondrial content)

Snap frozen soleus muscles were crushed under liquid nitrogen and placed in a cold Eppendorf tube with $20 \times$ ice cold Hepes buffer (containing 5 mM Hepes, 1 mM EGTA, 5 mM MgCl_2 , 1 mM DTT and 0.1% Triton X-100, pH 8), before being homogenised, sonicated and centrifuged. After centrifugation, 5 μl of supernatant, 250 μl of 0.1 M Tris, pH 8, 10 μl of 5,5'-dithiobis 2-nitrobenzoic acid (DTNB) and 1 μl acetyl coenzyme A were added to wells of a 96-well plate in duplicate. Ten microlitres of oxaloacetate was added to all wells to initiate the reaction and was assayed immediately. The absorbance of each well was then measured at 412 nm every 30 s for 5 min. Total protein content of each supernatant was measured using the detergent compatible (DC) protein assay (Bio-Rad, Gladesville, NSW, Australia). Enzyme activities were recorded as $\mu\text{mol min}^{-1} \text{mg protein}^{-1}$.

Protein extraction and immunoblotting (for muscle inflammation, mitochondrial activity and liver enzymes)

Frozen livers and gastrocnemius muscles were crushed under liquid nitrogen and homogenised in $10 \times$ ice-cold 1% NP40, 1 mM EDTA in PBS, supplemented with complete EDTA-free protease inhibitor tablets and PhosSTOP phosphatase inhibitor tablets (Roche Australia, Dee Why, NSW, Australia), and centrifuged (12 000 g, 10 min, 4°C). The protein concentration of supernatants was quantified using the DC protein assay (see above). Samples were resolved on 4–15%

SDS-PAGE TGX gels (Bio-Rad) and transferred onto nitrocellulose membrane using a Trans Turbo Blot system (Bio-Rad). Immunoblotting was performed with antibodies to neutrophil elastase (ab68672, Abcam, Cambridge, MA, USA), macrophage F4/80 (ab74383, Abcam), cysteine dioxygenase type 1 (ab53436, Abcam), cysteine sulfinate decarboxylase (ab101847, Abcam) and glyceraldehyde 3-phosphate dehydrogenase (GAPDH; 14C10, Cell Signaling Technology, Inc., Danvers, MA, USA), all dissolved 1:1000 in 5% bovine serum albumin. Horseradish peroxidase-conjugated secondary antibodies were from Thermo Fisher Scientific (Waltham, MA, USA). Chemiluminescence signal was captured using the ChemiDoc MP Imaging System (Bio-Rad). Resultant images were quantified using ImageJ software (Schneider *et al.* 2012). GAPDH loading controls were immunoblotted on the same membrane as the immunoblotted protein. All representative immunoblots in figures represent proteins immunoblotted on the same membrane as the loading control GAPDH.

Myeloperoxidase activity (to quantify activated inflammatory cells)

The enzyme myeloperoxidase (MPO), which is found in high levels in neutrophils and macrophages, catalyses the production of hypochlorous acid from hydrogen peroxide and chloride (Winterbourn & Kettle, 2000). Hypochlorous acid reacts with 2-[6-(4-aminophenoxy)-3-oxo-3H-xanthen-9-yl]benzoic acid (APF) to form the highly fluorescent compound fluorescein, which is measured in this method as described previously (Terrill *et al.* 2016). Briefly, frozen gastrocnemius muscles were crushed under liquid nitrogen and homogenised in 0.5% hexadecyltrimethylammonium bromide in PBS. Samples were centrifuged and supernatants were diluted in PBS. Human MPO was used as the standard for the assay (Cayman Chemical, Ann Harbor, MI, USA). Aliquots of each experimental sample or MPO standard were pipetted into a 384-well plate, before the addition of APF working solution (20 μM APF and 20 μM hydrogen peroxide in PBS). The plate was incubated at room temperature (protected from light) for 30 min, with fluorescence measured every minute using excitation at 485 nm and emission at 515–530 nm. The rate of change of fluorescence for each sample was compared to that of the standards and results were expressed per milligram of protein, quantified using the DC protein assay (Bio-Rad).

Protein thiol oxidation analysis

Reduced and oxidised protein thiols were measured in muscles using the 2 tag technique as described previously (Terrill *et al.* 2013a, 2016). In brief, frozen

EDL muscles were crushed under liquid nitrogen, before protein was extracted with 20% trichloroacetic acid (TCA) in acetone. Protein was solubilised in 0.5% SDS with 0.5 M Tris at pH 7.3 (SDS buffer) and protein thiols were labelled with the fluorescent dye BODIPY FL-*N*-(2-aminoethyl) maleimide (FLM, Invitrogen, Carlsbad, CA, USA). Following removal of the unbound dye using ethanol, protein was re-solubilised in SDS buffer, pH 7, and oxidised thiols were reduced with tris(2-carboxyethyl)phosphine (TCEP) before the subsequent unlabelled reduced thiols were labelled with a second fluorescent dye Texas Red C2-maleimide (Texas red, Invitrogen). The sample was washed in ethanol and re-suspended in SDS buffer. Samples were read using a fluorescence plate reader (Fluostar Optima, BMG Labtech, Ortenberg, Germany) with wavelengths set at excitation 485 nm, emission 520 nm for FLM and excitation 595 nm, and emission 610 nm for Texas red. FLM and Texas red standards were generated by reacting each dye with excess ovalbumin. Results were expressed per milligram of protein, quantified using the DC protein assay (Bio-Rad).

Reduced and oxidised thiols of myosin and actin proteins were quantified using one-dimensional SDS-PAGE. Labelled samples (remaining from the plate assay above) were diluted to equivalent protein concentrations. FLM and Texas red standards were generated by reacting each dye with excess BSA. FLM- and Texas red-labelled standards were combined and both the standards and the samples were diluted by the addition of sample buffer [125 mM Tris, pH 6.8, 4% SDS, 30% (v/v) glycerol, 0.02% bromophenol blue]. Standards and samples were applied to a 12% polyacrylamide gel. Gel electrophoresis was performed using the Bio-Rad Mini Protean III system. Each fluorescent gel was scanned using a Typhoon Trio scanner (GE Healthcare, Little Chalfont, UK) for fluorescence, with wavelengths set at excitation 485 nm, emission 520 nm for FLM and excitation 595 nm, and emission 610 nm for Texas red. The predicted myosin and actin bands were quantified by densitometry using ImageJ version 1.41 software (Schneider *et al.* 2012) using the integrated density function, after first removing the background. To assess the reversible protein thiol oxidation state of specific protein bands, the densitometry of in-gel standards for FLM and Texas red was used to construct a polynomial standard curve. The quantity of FLM and Texas red bound to a particular protein band was calculated from the standard curves.

Cysteine, GSH, taurine and NAC analysis

Cysteine, GSH and taurine contents of liver and gastrocnemius muscle were measured using reversed-phase HPLC as previously described (Terrill *et al.* 2013a, 2016). Frozen tissues were crushed using a mortar and pestle under liquid nitrogen and homogenised in 25 times

5% TCA, and after centrifugation (10 000 g, 5 min, 4°C) supernatants were removed and stored at -80°C before analysis. Analytes were separated using HPLC with fluorescence detection, with pre-column derivatisation with *o*-phthalaldehyde (OPA) and 2-mercaptoethanol (2ME). OPA reacts rapidly with amino acids and sulfhydryl groups to yield intensely fluorescent derivatives, and 2ME, a reducing agent, prevents the OPA reagent from oxidising. Supernatants were mixed with iodoacetamide (25 mM) in 5% TCA. An internal standard, *o*-phospho-DL-serine, dissolved in 5% TCA was added to a final concentration of 5 mM. Sodium borate was used to adjust the pH to 9. Samples were placed in an auto-sampler, which was maintained at 4°C. Samples were mixed on a sample loop with a derivatising solution containing 40 mM OPA and 160 mM 2ME in 100 mM sodium borate, pH 12, for 30 s before injection onto the column. Separation was achieved with a C18 column (5 μ l, 4.6 $\times$ 150 mm, Phenomenex, Torrance, CA, USA) using a Dionex Ultimate 3000 HPLC system. Mobile phase A consisted of 50 mM potassium phosphate buffer, methanol and tetrahydrofuran (94:3:3). Mobile phase B consisted of 90% methanol, with a gradient increase in B from 0 to 25%. Fluorescence was set at 360 nm and 455 nm for excitation and emission, respectively. Metabolite concentrations are expressed per milligram of protein. The protein content of muscle samples was quantified by solubilising the pellet in 0.5 M sodium hydroxide, before incubation at 80°C for 15 min. Once fully dissolved, protein concentrations of supernatants were quantified using a Bradford protein assay (Bio-Rad).

NAC in liver and muscle was also measured using reversed-phase HPLC, with some modifications to the above method. TCA-extracted supernatants (as above) were mixed with an internal standard, 2-hydroxyethyl disulfide (final concentration 1 mM), plus leucine and DTT (to final concentrations of 45 and 2.5 mM respectively). Sodium borate was used to adjust the pH to 8. Samples were placed in an autosampler, which was maintained at 4°C. Samples were mixed on a sample loop with a derivatising solution containing 25 mM OPA in 100 mM sodium borate, pH 10.4, for 30 s before injection onto the column. Separation and protein quantification were performed as above. The limit of quantification for NAC was quantified, and was shown to be 0.5 μ M, equating to approximately 50 pmol mg protein⁻¹, but no NAC was detected in either tissue (data not shown).

Statistical analysis

All experimental results are presented as means (standard deviation, SD), unless otherwise stated. The effects of NAC treatment on weekly measures of grip strength and body weight, and the effect of stimulation frequency on EDL force production were analysed using repeated-measures

two-way ANOVA. Where significant interactions were identified, subsequent *post hoc* analyses were carried out using repeated-measures one-way ANOVA. The effects of mouse strain and NAC treatment on dependent variables were analysed using two-way ANOVA or unpaired *t* test where appropriate using the statistical software package IBM SPSS Statistics (version 23, IBM, Armonk, NY, USA). Statistical significance was accepted at $P < 0.05$.

Results

Effect of NAC on growth of mice

Analysis of body weights in C57 and *mdx* mice across the 6 week treatment period revealed a significant increase in body weights ($P < 0.001$), significantly lower body weights in NAC-treated mice ($P < 0.001$), and a significant interaction between NAC treatment and time ($P < 0.001$). Whereas the body weight for untreated C57 and *mdx* mice increased significantly across all 6 weeks ($P < 0.001$), body weight gain was significantly impaired by NAC treatment in both strains (Fig. 1A and B). After 1 week of NAC treatment, body weights for both C57 ($P < 0.01$)

and *mdx* mice ($P < 0.05$) were significantly decreased ($\sim 9\%$) relative to their starting body weights; subsequent weight gain during the following 5 weeks was significantly impaired, most notably in *mdx* mice (Fig. 1A and B). NAC-treated C57 mice showed small but significant increases in body weight in weeks 2, 3 and 6, whereas *mdx* mice did not display any incremental increases in body weight throughout the treatment period. Consequently, the total body weight gain during the 6 week experimental period was notably lower in NAC-treated mice (~ 2.5 g) compared with untreated mice (~ 10 g) of both strains. Due to the significantly impaired body weight gain in NAC-treated mice, additional measures of bone lengths and organ weights (Fig. 1C and D) were made in C57 mice to ascertain whether there was reduced growth. There was no marked effect on overall body growth as shown by no significant differences in mean tibial length ($P = 0.058$), femur length ($P = 0.252$) or heart mass ($P = 0.128$) between NAC-treated and untreated C57 mice (Fig. 1C and D), although NAC significantly reduced ($\sim 17\%$) the C57 mean liver weight ($P = 0.014$; Fig. 1D). Furthermore, the EDL muscle weights in NAC-treated *mdx* (but not C57) mice were significantly lower ($\sim 22\%$) than untreated mice ($P = 0.006$; Fig. 1E).

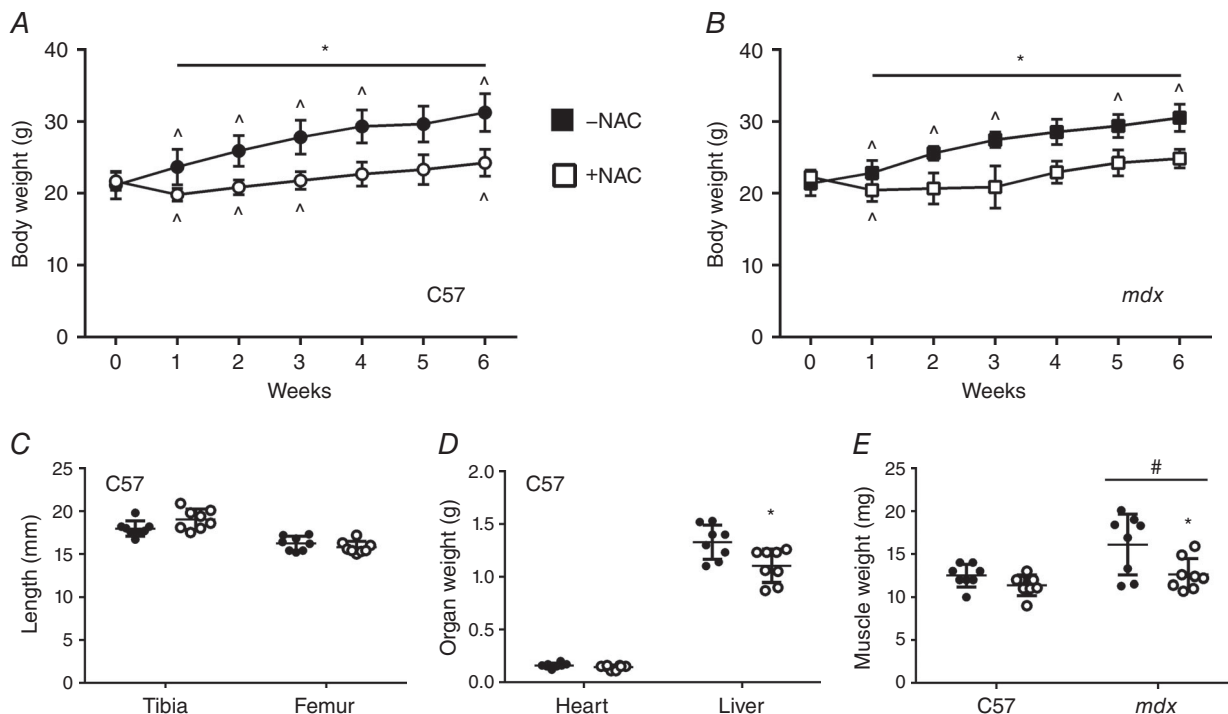


Figure 1. Effect of NAC on body weight (A, C57; B, *mdx*), bone length (C), organ weight (D) and muscle weight (E)

Body weights in NAC-treated mice (open symbols) were significantly lower compared with untreated mice (filled symbols) throughout the 6 week experiment. Liver weight in C57 mice and muscle weight in *mdx* mice were significantly lower following NAC treatment. Circles, C57; squares, *mdx* mice. Data are presented as mean (SD) in A and B and as individual values with horizontal lines indicating mean and SD in C–E ($n = 8$). *Body weight significantly different to preceding week, $P < 0.05$; *NAC significantly different to untreated mice, $P < 0.001$; #*mdx* significantly different to C57 mice, $P < 0.05$.

Effect of NAC on *in vivo* grip strength

Forelimb grip strength was measured weekly during the 6 week experiment and data are presented as absolute grip strength measured in grams of force and as grip strength normalised to body weight. In C57 mice, there was no significant effect of NAC on absolute grip strength ($P = 0.289$; Fig. 2A); however, when normalised to body weight, grip strength was significantly greater in NAC-treated, compared with untreated, C57 mice ($P < 0.001$; Fig. 2C). In *mdx* mice, NAC resulted in significantly greater absolute ($P = 0.001$) and normalised ($P < 0.001$) grip strength measurements, compared with untreated *mdx* mice (Fig. 2B and D). The increase in normalised grip strength (for C57 and *mdx* mice) was evident by 2 weeks of NAC treatment and persisted until the end of the 6 week experiment.

Effect of NAC treatment on *in vitro* muscle contractile function

Maximum specific force ($P < 0.001$) and total (non-normalised) peak tetanic force ($P = 0.002$) production were significantly greater for the isolated EDL

muscles of C57, compared with *mdx* mice (Fig. 3A and B). NAC treatment had no significant effect on the maximum specific force of EDL muscles from C57 mice ($P = 0.743$; Fig. 3A), whereas this was significantly increased by NAC treatment in *mdx* mice ($P = 0.038$; Fig. 3A). Importantly, there was no effect of NAC treatment on the total (non-normalised) peak tetanic force production for either strain (C57 $P = 0.418$; *mdx* $P = 0.593$; Fig. 3B), indicating that despite being smaller, the EDL muscles from NAC-treated *mdx* mice produced the same amount of force as untreated *mdx* mice.

Force–frequency relationships were recorded in EDL muscles from C57 (circles) and *mdx* (squares) mice to examine the effect of NAC across a range of submaximal stimulation intensities (Fig. 3C). Specific force in untreated *mdx* mice was significantly lower than C57 mice between stimulation frequencies of 30 and 150 Hz ($P = 0.001$), indicating that the dystrophic muscle weakness is not limited to maximal activation. There was no significant effect of NAC treatment on the force–frequency relationship in either strain.

There were no significant differences in any of the twitch contraction parameters (peak twitch force, time-to-peak, half-relaxation time, or maximum rate of

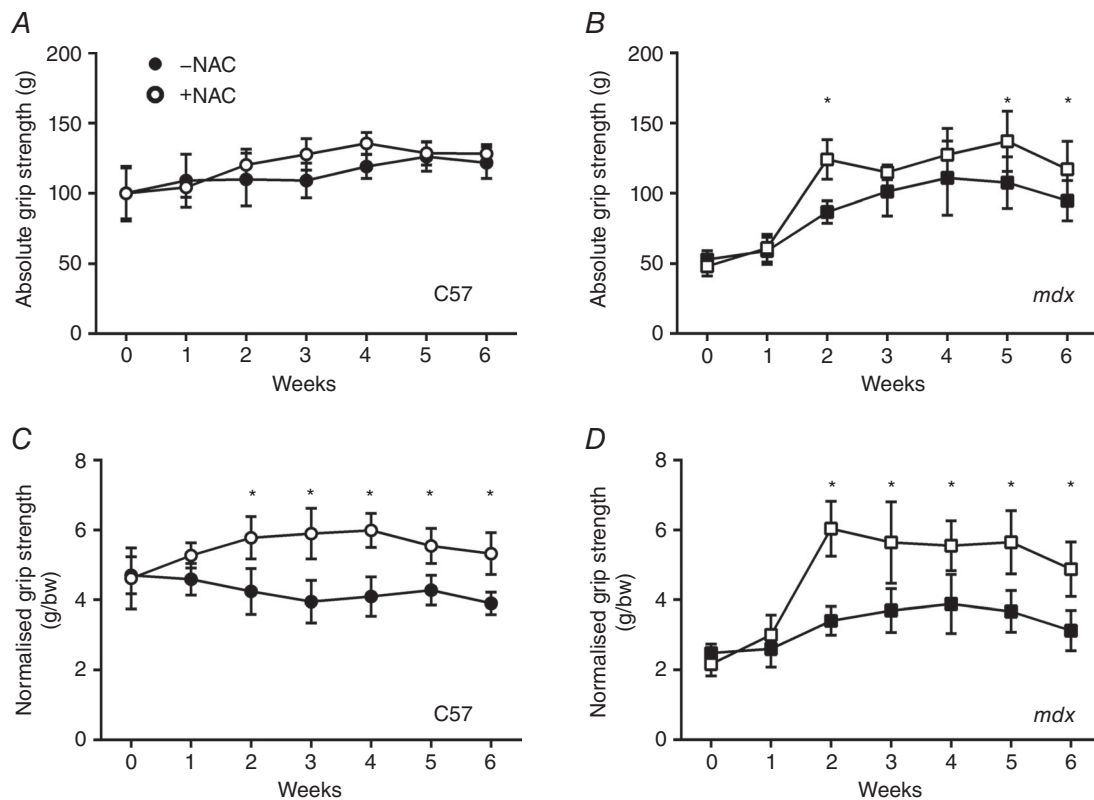


Figure 2. Effect of NAC on forelimb grip strength in C57 and *mdx* mice

A, C57 absolute grip strength; B, *mdx* absolute grip strength; C, C57 grip strength normalised to body weight; D, *mdx* grip strength normalised to body weight. Open symbols, NAC treated; closed symbols, un-treated; circles, C57; squares, *mdx* mice. Data are presented as mean (SD; $n = 8$). *NAC significantly different to untreated mice, $P \leq 0.001$.

force development) between untreated and NAC-treated groups for either C57 or *mdx* mice (Table 1). Fatigue resistance (FI), evaluated in isolated EDL muscles of untreated and NAC-treated C57 and *mdx* mice, was significantly greater in untreated *mdx*, compared with C57, mice ($P < 0.001$); however, NAC treatment had no significant effect on fatigue resistance or post-fatigue recovery of force in either strain ($P = 0.657$, $P = 0.608$, respectively; Table 1).

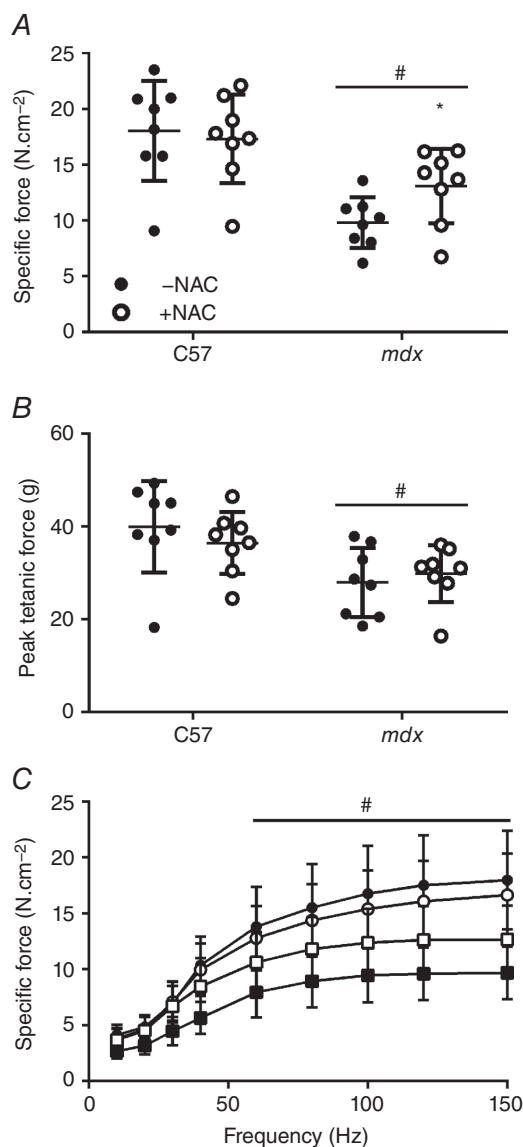


Figure 3. Effect of NAC on maximum specific force (A), peak tetanic force and (B) and force–frequency relationship (C) in isolated EDL muscles from C57 and *mdx* mice

Open symbols, NAC treated; closed symbols, un-treated; circles, C57; squares, *mdx* mice. Data are presented as individual values with horizontal lines indicating mean and SD in A and B and as mean and SD in C ($n = 8$). *NAC significantly different to untreated mice, $P = 0.038$; # *mdx* significantly different to C57 mice, $P < 0.01$.

Effect of NAC on citrate synthase in soleus muscles

To further investigate the improvements in muscle function following NAC treatment, citrate synthase activity (a measure of mitochondrial content; Holloszy *et al.* 1970) was analysed in soleus muscles from each group. Citrate synthase activity was not significantly different between untreated C57 and *mdx* mice [C57 = 5.2 (1.7); *mdx* = 3.5 (2.7) nmol min⁻¹ mg protein⁻¹; $P = 0.766$]. For C57 mice, there was no significant effect of NAC on citrate synthase activity [C57+NAC = 4.9 (2.3) nmol min⁻¹ mg protein⁻¹; $P = 0.785$]. In contrast, in *mdx* mice NAC significantly increased citrate synthase activity [*mdx*+NAC = 6.2 (1.5) nmol min⁻¹ mg protein⁻¹; $P = 0.007$] compared with untreated controls, indicating that NAC increased mitochondrial content in dystrophic *mdx* muscles.

Effect of NAC on inflammation in gastrocnemius muscles

Markers of skeletal muscle inflammation were examined in gastrocnemius muscles of all mice. In untreated mice, the neutrophil content (quantified by protein levels of neutrophil elastase) was significantly higher in *mdx* compared with C57 muscle ($P < 0.001$), and NAC significantly reduced this neutrophil content of *mdx* muscle ($P = 0.002$; Fig. 4A). Macrophage content (measured by levels of the protein F4/80) was also significantly higher in *mdx* compared with C57 muscle ($P = 0.001$), but was not affected by NAC treatment (C57 $P = 0.053$; *mdx* $P = 0.726$; Fig. 4B). The activity of MPO (a useful biomarker of inflammatory cells in tissues; Winterbourn *et al.* 2000) was significantly higher in *mdx* compared with C57 muscle ($P < 0.001$) and NAC treatment significantly reduced MPO activity in *mdx* muscle ($P = 0.016$), with no effect in C57 muscle ($P = 0.815$; Fig. 4C). These results confirm significantly elevated levels of inflammation, and high numbers of neutrophils and macrophages in *mdx* muscles, and additionally demonstrate that NAC treatment results in a significant reduction of some inflammatory markers in *mdx* muscles.

Effect of NAC on thiol oxidation in EDL muscles

The effect of NAC on oxidative stress in skeletal muscles was determined from analysis of protein thiol oxidation in isolated EDL muscles (Fig. 5). Total protein thiols were significantly lower in *mdx* compared with C57 muscles ($P < 0.001$; Fig. 5A); however, there was no significant effect of NAC treatment in either strain (C57 $P = 0.173$; *mdx* $P = 0.540$). There was no difference in the percentage of oxidised thiols between strains ($P = 0.704$), nor with NAC treatment ($P = 0.908$; Fig. 5B). For untreated mice,

Table 1. Effect of NAC on the mean ($\pm$ SD) peak twitch parameters and muscle fatigue in isolated EDL muscles from C57 and *mdx* mice ($n = 8$)

	C57		<i>mdx</i>	
	Untreated	NAC	Untreated	NAC
P_t (N cm $^{-2}$)	3.83 $\pm$ 1.03	3.63 $\pm$ 0.96	2.65 $\pm$ 0.61	2.80 $\pm$ 1.02
TTP (ms)	17.4 $\pm$ 1.3	17.3 $\pm$ 1.7	17.7 $\pm$ 1.7	17.9 $\pm$ 1.2
$\frac{1}{2}$ RT (ms)	24.5 $\pm$ 8.1	24.6 $\pm$ 4.9	20.8 $\pm$ 5.4	23.6 $\pm$ 7.0
dF/dt (g s $^{-1}$)	1993 $\pm$ 432	2060 $\pm$ 450	2193 $\pm$ 598	2527 $\pm$ 1614
Fatigue index	25.1 $\pm$ 5.2	24.9 $\pm$ 3.7	31.7 $\pm$ 5.6	33.5 $\pm$ 4.2 [#]
Fatigue recovery	85.0 $\pm$ 3.1	90.3 $\pm$ 5.4	91.2 $\pm$ 7.9	88.6 $\pm$ 10.9

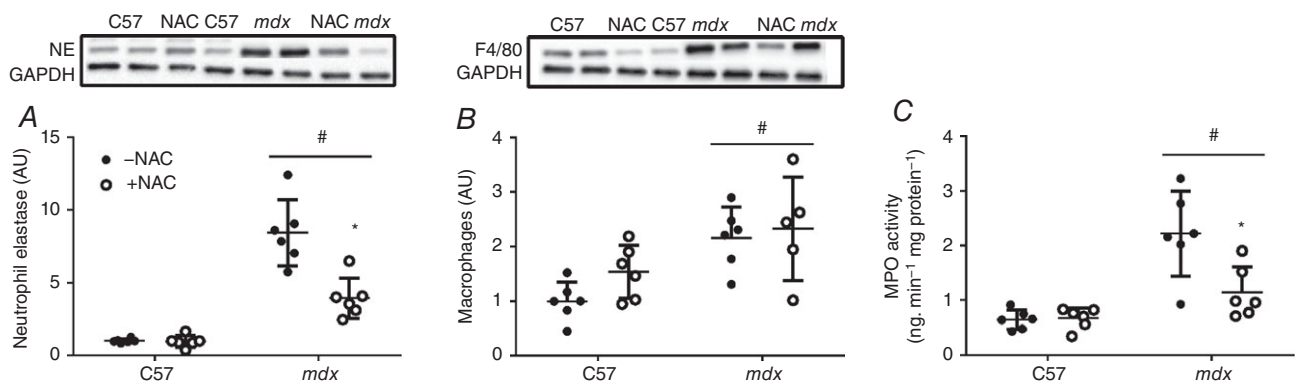
P_t , peak twitch force; TTP, time to peak twitch force; $\frac{1}{2}$ RT, half-relaxation time; dF/dt, maximum rate of force development. Fatigue index and fatigue recovery are presented as % of pre-fatigue force. [#]Significantly different to C57; $P < 0.001$.

there were no significant differences between strains for the levels of thiol oxidation of the actin ($P = 0.242$) and myosin ($P = 0.225$) proteins, and no effect of NAC on these measures in C57 mice ($P = 0.181$; Fig. 5C and D). However, NAC treatment significantly reduced myosin protein thiol oxidation in EDL muscles of *mdx* mice ($P = 0.036$; Fig. 5D).

Effect of NAC on cysteine metabolism in liver and muscle

We did not detect any NAC in livers or muscles of NAC-treated mice (the reasons for this are considered in the Discussion). However, the measurements of cysteine and GSH in these tissues provided insight into the mechanisms underlying the benefits and adverse effects of NAC treatment. For untreated mice, there was no significant difference between strains for the cysteine content of the liver or muscle ($P = 0.428$ and $P = 0.052$, respectively; Fig. 6A and E) nor for

the muscle GSH content ($P = 0.618$; Fig. 6F). GSH was significantly higher in the liver of *mdx* compared with C57 mice ($P < 0.001$; Fig. 6B); however, NAC treatment had no effect on any of these measures in either C57 or *mdx* tissues. Since we have shown previously that oral treatment with the cysteine-based compound L-2-oxothiazolidine-4-carboxylate (OTC) significantly increases liver and muscle taurine content in *mdx* mice (Terrill *et al.* 2013a), we also measured levels of taurine and the liver enzymes cysteine deoxygenase (CD) and cysteine sulfinatase decarboxylase (CSD) that are involved in the taurine synthesis pathway, to evaluate these pathways of NAC metabolism. Taurine content was significantly higher in the *mdx* liver ($P = 0.001$), but not *mdx* muscle ($P = 0.260$), compared with C57 mice but, again, there was no effect of NAC treatment (Fig. 6C and G). In *mdx* mice, levels of the liver enzyme CSD were significantly lower compared with C57 mice ($P < 0.001$), although levels of CD were not significantly different between strains ($P = 0.070$), and none of these enzyme levels

**Figure 4.** Effect of NAC on inflammation in gastrocnemius muscles from C57 and *mdx* mice

Quantification of protein levels for neutrophil elastase to identify neutrophils (A) and of F4/80 to identify macrophages (B) and measurement of the activity of the enzyme myeloperoxidase (MPO) to evaluate inflammatory cell activity (C). Open symbols, NAC treated; closed symbols, un-treated. Data are presented as individual values with horizontal lines indicating mean and SD ($n = 6$). *NAC significantly different to untreated mice, $P < 0.05$; [#]*mdx* significantly different to C57 mice, $P \leq 0.001$. Representative blots for neutrophil elastase and macrophages are shown; protein was standardised to GAPDH. AU, arbitrary units.

was significantly affected by NAC treatment (Fig. 6D and H).

Discussion

Overview

NAC has been proposed as a potential therapeutic treatment for DMD boys, with many benefits reported by several independent groups using dystrophic *mdx* mice. Consistent with previous observations, we provide evidence that treatment of male *mdx* mice with NAC (2% NAC in drinking water for 6 weeks) improves normalised measures of muscle function (with normalised grip strength increased within 2 weeks, and maximum specific force of EDL muscle increased by 25% at 12 weeks) and provide new insight into the mechanisms for these benefits of NAC. In the *mdx* mice sampled at 12 weeks of age, the NAC treatment resulted in many biochemical changes: increased mitochondrial content in the soleus muscle indicated by increased citrate synthase activity, reduced inflammation in the gastrocnemius muscle as demonstrated by decreased numbers of neutrophils and decreased MPO activity, and decreased myosin protein

thiol oxidation in the EDL. However, the most striking finding was that NAC treatment was associated with rapid reductions in body weight (within 1 week) and impaired weight gain during the treatment, along with significantly reduced weights of liver and EDL muscle at the end of the 6 week treatment. These reductions in body and organ weights have important implications for the interpretation of muscle function based on normalised force measures. Furthermore, impaired weight gain could be interpreted as deleterious, particularly in young growing mice, and has implications that need to be carefully considered in the context of the potential use of NAC as a treatment for DMD boys. These issues are discussed in detail below.

Pros and cons of NAC (and related molecules) as treatments for muscular dystrophy

We previously showed that oral NAC supplementation of adult *mdx* mice before treadmill running (at 12 weeks of age) greatly reduces the myofibre necrosis that is usually induced by exercise (Terrill *et al.* 2012). Similar reductions in myofibre necrosis were reported in diaphragm muscle of young *mdx* mice (aged 4 weeks) following daily

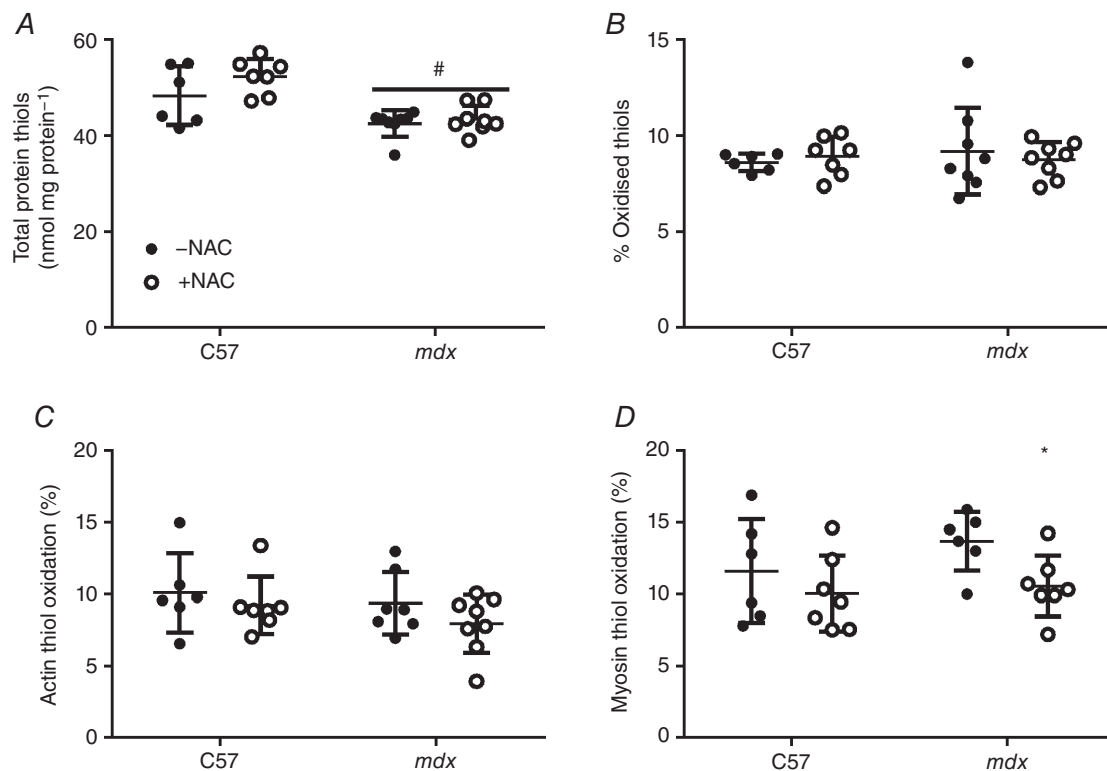


Figure 5. Effect of NAC on thiol oxidation in isolated EDL muscles from C57 mice and *mdx* mice

A, total protein thiols; B, percentage of oxidised thiols; C, percentage of actin thiol oxidation; D, percentage of myosin thiol oxidation. Open symbols, NAC-treated mice (C57, $n = 7$; *mdx*, $n = 8$), closed symbols, un-treated mice (C57, $n = 6$; *mdx*, $n = 7$). Data are presented as individual values with horizontal lines indicating mean and SD ($n = 8$). *NAC significantly different to untreated mice, $P = 0.022$; #*mdx* significantly different to C57 mice, $P < 0.001$.

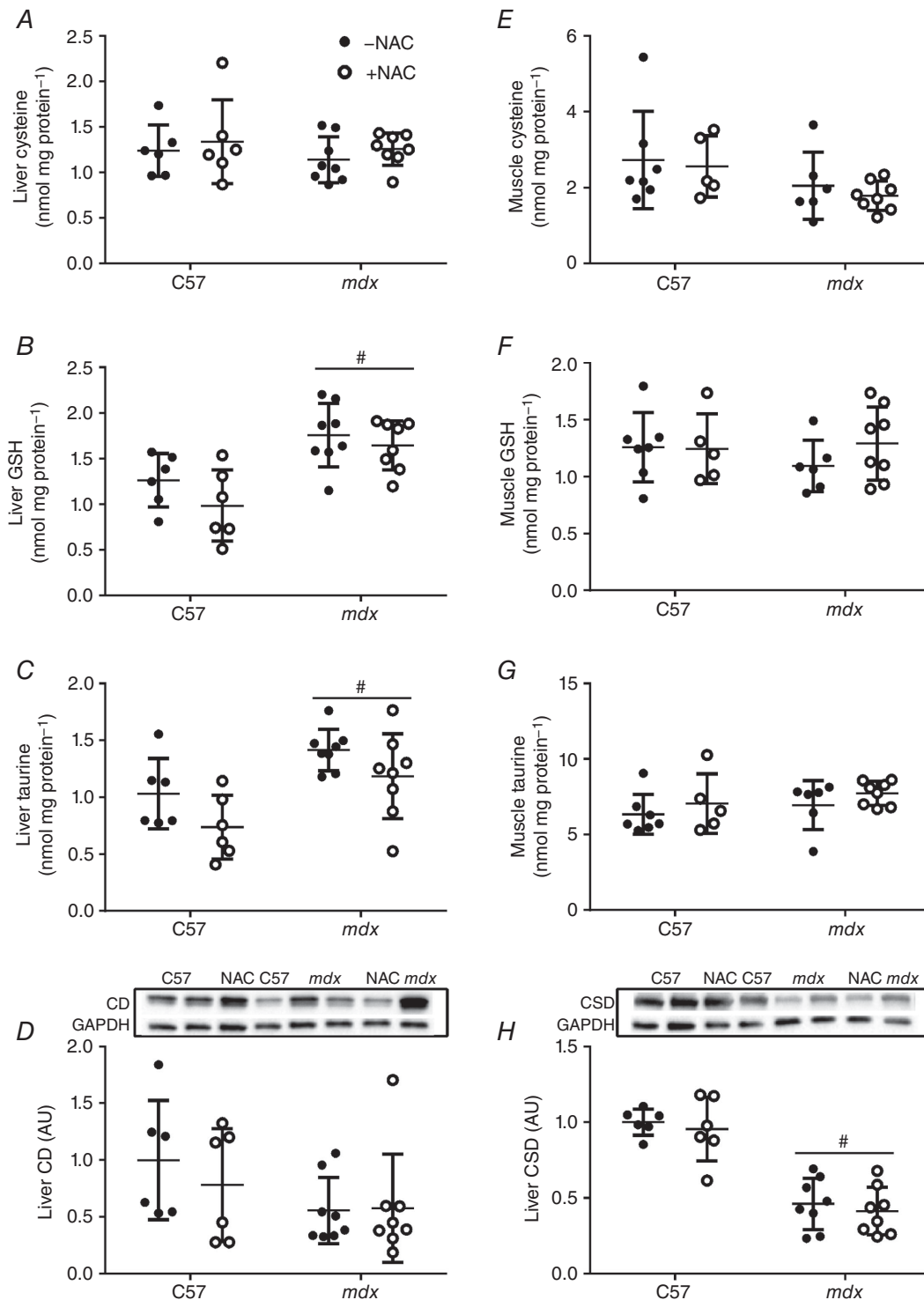


Figure 6. Effect of NAC on cysteine metabolism and liver enzymes in C57 and *mdx* mice

Cysteine (A, E), GSH (B, F) and taurine (C, G) concentrations in liver and gastrocnemius muscle, respectively; liver protein content of cysteine deoxygenase (CD, D) and cysteine sulfinate decarboxylase (CSD, H). Representative blots for CD and CSD are shown, and protein was standardised to GAPDH. AU, arbitrary units. Open symbols, NAC treated; filled symbols, un-treated mice. Data are presented as individual values with horizontal lines indicating mean and SD (for liver: C57 *n* = 6; *mdx* *n* = 8; for muscle C57-NAC *n* = 7, C57+NAC *n* = 5; *mdx*-NAC *n* = 6; *mdx*+NAC *n* = 8). #*mdx* significantly different to C57 mice, *P* < 0.001.

intraperitoneal injections of NAC for 2 weeks (de Senzi Moraes Pinto *et al.* 2013). Whitehead *et al.* (2008) were the first to report benefits of NAC, with significant *ex vivo* protection against eccentric contraction and stretch-induced damage of *mdx* muscle, as well as reduced oxidative stress, NF- κ B p65 and amount of myofibre damage (measured by Evans Blue Dye uptake), and decreased numbers of central myonuclei following *in vivo* supplementation with NAC (Whitehead *et al.* 2008). To our knowledge, the present study is the first to show that oral supplementation with NAC improves the function of *mdx* muscles *in vivo* (i.e. increased grip strength), as well as increasing maximum specific force *ex vivo* in isolated dystrophic muscle from *mdx* mice. The modest yet significant increase in maximum specific force after NAC treatment demonstrates specific improvement in the quality of dystrophic muscles, which was supported by increased citrate synthase activity, indicating increased mitochondrial content. However, it should be acknowledged that citrate synthase activity was measured in the slow-twitch soleus muscle and does not necessarily reflect increased activity in predominantly fast-twitch muscles such as the EDL. Decreased inflammation (decreases in neutrophils and MPO activity in the gastrocnemius muscle) and protein thiol oxidation (in EDL) provide further evidence of the beneficial effects of NAC on the quality of dystrophic muscle. We acknowledged that although *mdx* mice provide a useful model for DMD, they display a less severe phenotype than the human condition. It would be important to establish if similar beneficial effects of NAC treatment on dystrophic muscle function are evident in more severe models of human DMD such as the dystrophin–utrophin double-knockout mouse (*mdx/utrn*^{-/-}) (Deconinck *et al.* 1997) or the golden retriever muscular dystrophy (GRMD) dog model (Sharp *et al.* 1992).

These results are consistent with the hypothesis that exogenous antioxidants such as NAC can attenuate the inflammation and oxidative stress that contribute to contractile dysfunction. Previous studies using NAC treatment have also reported reduced levels of protein thiol oxidation (Matuszczak *et al.* 2005; Terrill *et al.* 2012), oxidative stress (Whitehead *et al.* 2008), tumour necrosis factor (de Senzi Moraes Pinto *et al.* 2013) and NF- κ B signalling (Farid *et al.* 2005; Whitehead *et al.* 2008). However, it is currently unclear whether NAC acts directly as an ROS scavenger, or indirectly via the synthesis of endogenous antioxidants or other cysteine derivatives such as taurine.

We attempted to measure the NAC concentrations in liver and muscle, but did not detect any within the limits of quantification (50 pmol mg protein⁻¹). This lack of NAC detection is probably not surprising given the short plasma half-life (1.95 h) and low oral bioavailability (4%) of NAC in the reduced (compared with oxidised) form

(Olsson *et al.* 1988). Systemically, NAC is converted to cysteine, which can be toxic at high levels and significantly impairs growth (Dilger *et al.* 2007). In support of this, we have shown that NAC-treated mice had significantly lower body weights (in *mdx* and C57 mice), liver weight (C57 mice) and muscle weights (*mdx* mice), which could arise from cysteine toxicity. To test this hypothesis, we measured cysteine levels in the liver and skeletal muscles, but these parameters were not affected by NAC treatment in either *mdx* or C57 mice. Toxic accumulation of cysteine may be prevented through the synthesis of GSH or another downstream product, taurine, via the actions of the enzymes CD and CSD (Terrill *et al.* 2015). Surprisingly, NAC treatment had no significant effect on either CD or CSD content, nor on the liver or muscle concentrations of GSH. It is possible that NAC treatment transiently increased taurine synthesis that was ultimately excreted, due to a down-regulation of taurine reabsorption in the renal tubules, a process that we have shown is already dysregulated in *mdx* kidney (Terrill *et al.* 2015). Alternative pathways for cysteine disposal, such as transamination to sulfate (Huxtable, 2000), may also be activated. These findings do not provide support for NAC-induced cysteine toxicity and alternative mechanisms underlying the negative effects on growth should be investigated. Further research is clearly warranted to clarify the metabolism and systemic regulation of NAC and similar cysteine precursors, especially in the context of young growing mammals.

We have recently evaluated alternative therapeutic compounds in *mdx* mice including OTC and taurine (Terrill *et al.* 2013a, 2016). Oral administration of the cysteine precursor OTC (0.5%, w/v) to *mdx* mice from 6 to 12 weeks of age produced similar improvements in grip strength, and reductions in MPO activity and protein thiol oxidation (Terrill *et al.* 2013a), as observed with NAC treatment over the same ages in the present study. Importantly, OTC treatment had no effect on body weight nor muscle weight in these adult *mdx* mice (Terrill *et al.* 2013a), suggesting that OTC is a superior alternative treatment to NAC. Yet, when OTC was administered to very young growing *mdx* mice from weaning (18 days) to 6 weeks of age, the improvements in muscle function were compromised by reduced body and muscle weights (Terrill *et al.* 2016), similar to that which we report here with NAC treatment of the more mature *mdx* mice. In contrast, oral administration of taurine during this critical growth period significantly improved muscle function, and reduced inflammation and protein thiol oxidation without any detrimental effects on growth parameters (Terrill *et al.* 2016). These findings suggest that taurine supplementation may be superior to NAC or OTC as a therapeutic treatment for DMD boys. Indeed, independent studies from at least three groups now strongly support the benefits of taurine for reduced

dystrotophology in animal models of DMD (De Luca *et al.* 2015; Horvath *et al.* 2016; Terrill *et al.* 2016).

The most notable benefits of NAC treatment in *mdx* mice were the significant increases in normalised grip strength and maximum specific force of EDL muscles. However, these outcome measures are strongly influenced by body weight and muscle weights, respectively, which were both significantly reduced in NAC-treated mice. Therefore, interpretation about muscle function based on normalised force measures should be made with caution, especially when the treatment in question influences the normalisation parameters. The lower body weights of NAC-treated C57 and *mdx* mice could be considered deleterious and thus raise concerns about the suitability of NAC as a treatment for DMD boys, since these children and young men would require long-term treatment over decades, starting from a young age. Dosage, route of administration and age at onset of delivery are important factors to consider, especially in the context of very young growing boys with DMD. Our present experiment used 2% NAC for 6 weeks, whereas our previous study (Terrill *et al.* 2012) used only 1% NAC for 6 weeks (also starting at 6 weeks of age and sampling at 12 weeks), plus a short term high dosage regime of 4% NAC for 1 week before exercise. Whitehead *et al.* (2008) also used 1% NAC for 6 weeks, starting at 3 weeks of age during the rapid growth phase and de Senzi Moraes Pinto *et al.* (2013) also used very young *mdx* mice treated from 2 weeks of age with daily i.p. injections of NAC (150 mg kg⁻¹) for 2 weeks. Unfortunately, there are limited data available on the effects of NAC treatment on body weight or other phenotype parameters in these other studies and thus potentially adverse and/or indirect effects may not be readily identified. Although de Senzi Moraes Pinto *et al.* (2013) reported that NAC treatment had no effect on body weight, their study was of shorter duration (2 weeks) and used a different route of administration (i.p. versus oral) compared with our present study. Note that NAC is readily oxidised and is unstable in drinking water, and in humans oral ingestion of 150 mg kg⁻¹ NAC caused mild adverse reactions including dysphoria and gastrointestinal disturbances such as nausea and diarrhoea (Matuszczak *et al.* 2005). Similar gastrointestinal disturbances may have contributed to the lower body weights (and tissue weights) that we observed in the present study for both C57 and *mdx* mice and therefore may reflect indirect consequences of NAC treatment rather than direct effects on growth.

Adverse drug reactions have been widely reported in human studies using NAC to treat acetaminophen overdose (Miller & Rumack, 1983; Zyoud *et al.* 2010). While the majority of cases related to mild gastrointestinal reactions, the incidence and severity of reactions increased with the dose and duration of treatment. Similar reactions have been reported in healthy adults receiving chronic low dose

NAC (5.6 g day⁻¹ for 6 months) (Hermann, 1970) or acute high dose NAC (single 140 mg kg⁻¹ dose) (Ferreira *et al.* 2011). In addition to mild gastrointestinal reactions, NAC has been related to more severe side effects. For example, chronic systemic administration of NAC (for 3 weeks at 10 mg ml⁻¹) caused pulmonary arterial hypertension in mice (Palmer *et al.* 2007) which was attributed to NAC conversion to S-nitroso-N-acetylcysteine (SNOAC) and hypoxic signalling *in vivo*. Continuous i.v. infusion of high dose NAC (550 and 950 mg NAC kg⁻¹ in 48 h) during an lipopolysaccharide toxicity challenge in rats decreased lung GSH and increased mortality compared with rats receiving lipopolysaccharide alone (Sprong *et al.* 1998). In humans, NAC treatment (10 mg kg⁻¹) for 7 days after eccentric exercise-induced muscle damage significantly increased tissue damage and oxidative stress compared with untreated participants (Childs *et al.* 2001). Collectively, these studies highlight the potential for severe side effects arising from the long term administration of NAC.

Although oxidative stress is implicated in the pathogenesis of DMD (Disatnik *et al.* 1998; Rando *et al.* 1998), and may also occur as a consequence of myofibre necrosis and inflammation (Morgan *et al.* 2008), clinical trials of antioxidant therapies have been rather disappointing (Rando, 2002). NAC is a non-specific antioxidant with effects on multiple biological processes (Zafarullah *et al.* 2003). Due to the potential adverse effects of NAC on various parts of the body, additional investigation of the appropriate dose and route of administration in more severe animal models of DMD should be conducted before considering the use of NAC as a therapeutic treatment for DMD boys.

References

- Andrade FH, Reid MB, Allen DG & Westerblad H (1998). Effect of hydrogen peroxide and dithiothreitol on contractile function of single skeletal muscle fibres from the mouse. *J Physiol* **509**, 565–575.
- Andrade FH, Reid MB & Westerblad H (2001). Contractile response of skeletal muscle to low peroxide concentrations: myofibrillar calcium sensitivity as a likely target for redox-modulation. *FASEB J* **15**, 309–311.
- Arthur PG, Grounds MD & Shavlakadze T (2008). Oxidative stress as a therapeutic target during muscle wasting: considering the complex interactions. *Curr Opin Clin Nutr Metab Care* **11**, 408–416.
- Aruoma OI, Halliwell B, Hoey BM & Butler J (1989). The antioxidant action of N-acetylcysteine: its reaction with hydrogen peroxide, hydroxyl radical, superoxide, and hypochlorous acid. *Free Radic Biol Med* **6**, 593–597.
- Brooks SV & Faulkner JA (1988). Contractile properties of skeletal muscles from young, adult and aged mice. *J Physiol* **404**, 71–82.

- Bushby K, Finkel R, Birnkrant DJ, Case LE, Clemens PR, Cripe L, Kaul A, Kinnett K, McDonald C, Pandya S, Poysky J, Shapiro F, Tomezsko J & Constantin C (2010). Diagnosis and management of Duchenne muscular dystrophy, part 1: diagnosis, and pharmacological and psychosocial management. *Lancet Neurol* **9**, 77–93.
- Childs A, Jacobs C, Kaminski T, Halliwell B & Leeuwenburgh C (2001). Supplementation with vitamin C and N-acetyl-cysteine increases oxidative stress in humans after an acute muscle injury induced by eccentric exercise. *Free Radic Biol Med* **31**, 745–753.
- Crowder MS & Cooke R (1984). The effect of myosin sulphhydryl modification on the mechanics of fibre contraction. *J Muscle Res Cell Motil* **5**, 131–146.
- De Luca A, Pierro S & Camerino DC (2015). Taurine: the appeal of a safe amino acid for skeletal muscle disorders. *J Transl Med* **13**, 243.
- de Senzi Moraes Pinto R, Ferretti R, Moraes LH, Neto HS, Marques MJ & Minatel E (2013). N-Acetylcysteine treatment reduces TNF- α levels and myonecrosis in diaphragm muscle of *mdx* mice. *Clin Nutr* **32**, 472–475.
- Deconinck AE, Rafael JA, Skinner JA, Brown SC, Potter AC, Metzinger L, Watt DJ, Dickson JG, Tinsley JM & Davies KE (1997). Utrophin-dystrophin-deficient mice as a model for Duchenne muscular dystrophy. *Cell* **90**, 717–727.
- Dilger RN & Baker DH (2007). Oral N-acetyl-L-cysteine is a safe and effective precursor of cysteine. *J Anim Sci* **85**, 1712–1718.
- Dilger RN, Toue S, Kimura T, Sakai R & Baker DH (2007). Excess dietary L-cysteine, but not L-cystine, is lethal for chicks but not for rats or pigs. *J Nutr* **137**, 331–338.
- Disatnik MH, Dhawan J, Yu Y, Beal MF, Whirl MM, Franco AA & Rando TA (1998). Evidence of oxidative stress in *mdx* mouse muscle: studies of the pre-necrotic state. *J Neurol Sci* **161**, 77–84.
- Emery AE (2002). The muscular dystrophies. *Lancet* **359**, 687–695.
- Farid M, Reid MB, Li Y-P, Gerken E & Durham WJ (2005). Effects of dietary curcumin or N-acetylcysteine on NF- κ B activity and contractile performance in ambulatory and unloaded murine soleus. *Nutr Metab* **2**, 20.
- Fauconnier J, Thireau J, Reiken S, Cassan C, Richard S, Matecki S, Marks AR & Lacampagne A (2010). Leaky RyR2 trigger ventricular arrhythmias in Duchenne muscular dystrophy. *Proc Natl Acad Sci USA* **107**, 1559–1564.
- Ferreira LF, Campbell KS & Reid MB (2011). N-Acetylcysteine in handgrip exercise: plasma thiols and adverse reactions. *Int J Sport Nutr Exerc Metab* **21**, 146–154.
- Ferreira LF & Reid MB (2008). Muscle-derived ROS and thiol regulation in muscle fatigue. *J Appl Physiol* **104**, 853–860.
- Grounds MD (2008). Two-tiered hypotheses for Duchenne muscular dystrophy. *Cell Mol Life Sci* **65**, 1621–1625.
- Grundy D (2015). Principles and standards for reporting animal experiments in *The Journal of Physiology and Experimental Physiology*. *J Physiol* **593**, 2547–2549.
- Hauser E, Hoyer H, Bittner R, Widhalm K, Herkner K & Lubec G (1995). Oxyradical damage and mitochondrial enzyme activities in the *mdx* mouse. *Neuropediatrics* **26**, 260–262.
- Haycock JW, MacNeil S, Jones P, Harris JB & Mantle D (1996). Oxidative damage to muscle protein in Duchenne muscular dystrophy. *Neuroreport* **8**, 357–361.
- Hermann HW (1970). Chronic human toxicity study of orally administered acetylcysteine. *Mead Johnson Research Center Report (Herm-HW-03637) Final Study Report*.
- Holloszy JO, Oscai LB, Don IJ & Mole PA (1970). Mitochondrial citric acid cycle and related enzymes: adaptive response to exercise. *Biochem Biophys Res Commun* **40**, 1368–1373.
- Horvath DM, Murphy RM, Mollica JP, Hayes A & Goodman CA (2016). The effect of taurine and β -alanine supplementation on taurine transporter protein and fatigue resistance in skeletal muscle from *mdx* mice. *Amino Acids* **48**, 2635–2645.
- Huxtable RJ (2000). Expanding the circle 1975–1999: sulfur biochemistry and insights on the biological functions of taurine. *Adv Exp Med Biol* **483**.
- Jackson MJ (2008). Redox regulation of skeletal muscle. *IUBMB Life* **60**, 497–501.
- Kharraz Y, Guerra J, Pessina P, Serrano AL & Munoz-Canoves P (2014). Understanding the process of fibrosis in Duchenne muscular dystrophy. *Biomed Res Int* **2014**, 965631, <https://doi.org/10.1155/2014/965631>.
- Kim JH, Kwak HB, Thompson LV & Lawler JM (2013). Contribution of oxidative stress to pathology in diaphragm and limb muscles with Duchenne muscular dystrophy. *J Muscle Res Cell Motil* **34**, 1–13.
- Matthews E, Brassington R, Kuntzer T, Jichi F & Manzur AY (2016). Corticosteroids for the treatment of Duchenne muscular dystrophy. *Cochrane Database Syst Rev* CD003725.
- Matuszczak Y, Farid M, Jones J, Lansdowne S, Smith MA, Taylor AA & Reid MB (2005). Effects of N-acetylcysteine on glutathione oxidation and fatigue during handgrip exercise. *Muscle Nerve* **32**, 633–638.
- McKenna MJ, Medved I, Goodman CA, Brown MJ, Bjorksten AR, Murphy KT, Petersen AC, Sostaric S & Gong X (2006). N-acetylcysteine attenuates the decline in muscle Na⁺, K⁺-pump activity and delays fatigue during prolonged exercise in humans. *J Physiol* **576**, 279–288.
- Medved I, Brown MJ, Bjorksten AR, Murphy KT, Petersen AC, Sostaric S, Gong X & McKenna MJ (2004). N-Acetylcysteine enhances muscle cysteine and glutathione availability and attenuates fatigue during prolonged exercise in endurance-trained individuals. *J Appl Physiol* **97**, 1477–1485.
- Mendez J & Keys A (1960). Density and composition of mammalian muscle. *Metab Clin Exp* **9**, 184–188.
- Miller LF & Rumack BH (1983). Clinical safety of high oral doses of acetylcysteine. *Semin Oncol* **10**, 76–85.
- Mishima T, Yamada T, Matsunaga S & Wada M (2005). N-Acetylcysteine fails to modulate the in vitro function of sarcoplasmic reticulum of diaphragm in the final phase of fatigue. *Acta Physiol Scand* **184**, 195–202.
- Miyatake S, Shimizu-Motohashi Y, Takeda S & Aoki Y (2016). Anti-inflammatory drugs for Duchenne muscular dystrophy: focus on skeletal muscle-releasing factors. *Drug Des Devel Ther* **10**, 2745–2758.
- Morgan MJ, Kim YS & Liu ZG (2008). TNF α and reactive oxygen species in necrotic cell death. *Cell Res* **18**, 343–349.
- Olsson B, Johansson M, Gabrielsson J & Bolme P (1988). Pharmacokinetics and bioavailability of reduced and oxidized N-acetylcysteine. *Eur J Clin Pharmacol* **34**, 77–82.

- Palmer LA, Doctor A, Chhabra P, Sheram ML, Laubach VE, Karlinsey MZ, Forbes MS, Macdonald T & Gaston B (2007). S-Nitrosothiols signal hypoxia-mimetic vascular pathology. *J Clin Invest* **117**, 2592–2601.
- Petrof BJ, Shrager JB, Stedman HH, Kelly AM & Sweeney HL (1993). Dystrophin protects the sarcolemma from stresses developed during muscle-contraction. *Proc Natl Acad Sci USA* **90**, 3710–3714.
- Rando TA (2001). The dystrophin-glycoprotein complex, cellular signaling, and the regulation of cell survival in the muscular dystrophies. *Muscle Nerve* **24**, 1575–1594.
- Rando TA (2002). Oxidative stress and the pathogenesis of muscular dystrophies. *Am J Phys Med Rehabil* **81**, S175–186.
- Rando TA, Disatnik MH, Yu Y & Franco A (1998). Muscle cells from *mdx* mice have an increased susceptibility to oxidative stress. *Neuromuscul Disord* **8**, 14–21.
- Schneider CA, Rasband WS & Eliceiri KW (2012). NIH Image to ImageJ: 25 years of image analysis. *Nat Methods* **9**, 671–675.
- Sharp NJH, Kornegay JN, Vancamp SD, Herbstreith MH, Secore SL, Kettle S, Hung WY, Constantinou CD, Dykstra MJ, Roses AD & Bartlett RJ (1992). An error in dystrophin messenger-RNA processing in golden retriever muscular-dystrophy, an animal homolog of Duchenne muscular-dystrophy. *Genomics* **13**, 115–121.
- Shindoh C, DiMarco A, Thomas A, Manubay P & Supinski G (1990). Effect of N-acetylcysteine on diaphragm fatigue. *J Appl Physiol* **68**, 2107–2113.
- Shkryl VM, Martins AS, Ullrich ND, Nowycky MC, Niggli E & Shirokova N (2009). Reciprocal amplification of ROS and Ca²⁺ signals in stressed *mdx* dystrophic skeletal muscle fibers. *Pflugers Arch* **458**, 915–928.
- Sprong RC, Winkelhuyzen-Janssen AML, Aarsman CJM, van Oirschot JFLM, van der Bruggen T & van Asbeck BS (1998). Low-dose N-acetylcysteine protects rats against endotoxin-mediated oxidative stress, but high-dose increases mortality. *Am J Resp Crit Care* **157**, 1283–1293.
- Terrill JR, Boyatzis A, Grounds MD & Arthur PG (2013a). Treatment with the cysteine precursor l-2-oxothiazolidine-4-carboxylate (OTC) implicates taurine deficiency in severity of dystropathology in *mdx* mice. *Int J Biochem Cell Biol* **45**, 2097–2108.
- Terrill JR, Grounds MD & Arthur PG (2015). Taurine deficiency, synthesis and transport in the *mdx* mouse model for Duchenne Muscular Dystrophy. *Int J Biochem Cell Biol* **66**, 141–148.
- Terrill JR, Pinniger GJ, Graves JA, Grounds MD & Arthur PG (2016). Increasing taurine intake and taurine synthesis improves skeletal muscle function in the *mdx* mouse model for Duchenne muscular dystrophy. *J Physiol* **594**, 3095–3110.
- Terrill JR, Radley-Crabb HG, Grounds MD & Arthur PG (2012). N-Acetylcysteine treatment of dystrophic *mdx* mice results in protein thiol modifications and inhibition of exercise induced myofibre necrosis. *Neuromuscul Disord* **22**, 427–434.
- Terrill JR, Radley-Crabb HG, Iwasaki T, Lemckert FA, Arthur PG & Grounds MD (2013b). Oxidative stress and pathology in muscular dystrophies: focus on protein thiol oxidation and dysferlinopathies. *FEBS J* **280**, 4149–4164.
- Whitehead NP, Pham C, Gervasio OL & Allen DG (2008). N-Acetylcysteine ameliorates skeletal muscle pathophysiology in *mdx* mice. *J Physiol* **586**, 2003–2014.
- Whitehead NP, Yeung EW & Allen DG (2006). Muscle damage in *mdx* (dystrophic) mice: role of calcium and reactive oxygen species. *Clin Exp Pharmacol Physiol* **33**, 657–662.
- Williams IA & Allen DG (2007). The role of reactive oxygen species in the hearts of dystrophin-deficient *mdx* mice. *Am J Physiol Heart Circ Physiol* **293**, H1969–1977.
- Winterbourn CC & Kettle AJ (2000). Biomarkers of myeloperoxidase-derived hypochlorous acid. *Free Radic Biol Med* **29**, 403–409.
- Winterbourn CC, Vissers MC & Kettle AJ (2000). Myeloperoxidase. *Curr Opin Hematol* **7**, 53–58.
- Wood CL, Straub V, Guglieri M, Bushby K & Cheetham T (2016). Short stature and pubertal delay in Duchenne muscular dystrophy. *Arch Dis Child* **101**, 101–106.
- Zafarullah M, Li WQ, Sylvester J & Ahmad M (2003). Molecular mechanisms of N-acetylcysteine actions. *Cell Mol Life Sci* **60**, 6–20.
- Zyoud SH, Awang R, Sulaiman SAS, Sweileh WM & Al-jabi SW (2010). Incidence of adverse drug reactions induced by N-acetylcysteine in patients with acetaminophen overdose. *Hum Exp Toxicol* **29**, 153–160.

Additional information

Competing interests

All authors have no financial or personal conflict with other people or organisations that could inappropriately influence our work.

Author contributions

Experiments were performed at the University of Western Australia. GJP, JRT, EBA, MDG and PGA were responsible for the study design. EBA, JRT and GJP were involved in the acquisition and analysis of the data. All authors contributed to the interpretation of the data. EBA and GJP drafted the article. All authors revised the article critically for important intellectual content, have approved the final version of the manuscript and agree to be accountable for all aspects of the work. All persons designated as authors qualify for authorship.

Funding

This research was made possible by grant funding from the National Health and Medical Research Council (NHMRC) of Australia and the USA Parent Project for Muscular Dystrophy.